Literature Review

Streaks, Beliefs, and Biases: The Gambler's Fallacy and Hot Hand Effect

Introduction and Background

Judgments about streaks, runs of identical outcomes in sequential events, occupy a central place in research on human reasoning under uncertainty. From casino gambling and sports performance to financial markets and algorithmic decision systems, people routinely draw strong conclusions from patterns that normative probability theory treats as uninformative. Two of the most extensively documented tendencies in this domain are the gambler's fallacy, the expectation that a streak is likely to reverse, and the hot-hand belief, the expectation that a streak will continue. Recent research increasingly challenges the view that streak judgments reflect simple probabilistic error. Instead, evidence suggests that people engage in structured causal reasoning about the processes that generate outcomes. Foundational work shows that individuals struggle with statistical independence and expect short sequences to reflect long-run properties, a tendency rooted in intuitions about randomness rather than ignorance of probability theory (Tyszka et al., 2008; Oskarsson et al., 2009). Field and laboratory studies further demonstrate that expectations of reversal and continuation can coexist within the same individual and environment, varying with context, framing, and perceived source of outcomes (Croson & Sundali, 2005; Bar-Eli et al., 2006; Polin & Benisaac, 2023). Together, these findings motivate a shift away from cataloguing fallacies toward understanding how people infer structure, dependence, and change in uncertain environments.

This review synthesizes ~50 peer-reviewed articles spanning psychology, behavioural economics, and decision science to develop an integrated account of streak perception. It focuses primarily on work published over the past two decades, while drawing selectively on earlier foundational contributions. Three core themes emerge that form the structure of this review. First, expressions of the gambler's fallacy and hot-hand belief are highly context-dependent, reflecting flexible judgment rather than stable bias. Second, this flexibility is driven by beliefs about the generative structure of outcomes, specifically, whether processes are perceived as independent, state-dependent, or nonstationary. Third, attribution of agency and intentionality, including through anthropomorphism, operates as a higher-order cue that systematically shifts these beliefs by introducing assumptions of memory, learning, and control. By integrating classic work on randomness with contemporary research on agency attribution, this review aims to clarify how people interpret streaks and why identical sequences can give rise to sharply different predictions. Together, this synthesis helps clarify how people make sense of repeated outcomes in environments where randomness, structure, and agency are ambiguous, and can aid future work on sequential judgment in domains ranging from gambling and finance to human-AI interaction and trust in algorithmic systems.

Methodology

The review draws from systematic searches across major academic databases (Google Scholar, JSTOR, ScienceDirect, ResearchGate, and UofT Libraries) using keywords such as *gambler's fallacy, hot hand, streaks, random sequences, agency, and anthropomorphism*. Articles were screened based on relevance to psychological evaluations of sequential outcomes, resulting in the inclusion of ~50 studies. These were grouped into thematic clusters and additional literature, from which dominant concepts and relationships were extracted for synthesis.

Thematic Review and Synthesis

Theme 1: Context-Dependent Expression of Gambler's Fallacy and Hot-Hand Beliefs

A central insight emerging from the literature is that the gambler's fallacy and the hot-hand belief are not mutually exclusive cognitive errors, but context-dependent expressions of how individuals interpret streaks. Rather than committing to one bias uniformly, people flexibly shift between expectations of reversal and continuation depending on how the task environment is construed. This flexibility challenges early interpretations that treated negative and positive recency as stable individual differences or symmetric fallacies.

Foundational work on randomness perception provides the starting point for this argument. Bar-Hillel and Wagenaar (1991) demonstrate that people expect short sequences to resemble long-run distributions, a tendency labeled the "law of small numbers." This expectation produces negative recency when outcomes are framed as independent draws from a stationary random process. Field evidence from gambling environments reinforces this pattern: lottery players systematically avoid recently drawn numbers (Suetens et al., 2016), and this avoidance intensifies when the informational sample is small or perceived as insufficient (Wang & Li, 2025). Importantly, these findings do not imply superstition or confusion about probability; rather, they suggest that people implicitly expect random systems to self-correct locally.

At the same time, numerous studies demonstrate that this expectation is fragile and readily overturned. Xu and Harvey (2014) show that in online gambling markets, negative recency at the level of beliefs about outcomes can generate apparent hot-hand betting at the behavioural level, depending on how streaks are operationalized. Kong et al. (2020) provide more direct evidence of coexistence, showing that individuals simultaneously endorse gambler's fallacy reasoning for mechanical devices while applying hot-hand reasoning to human performance. Ayton and Fischer (2004) similarly argue that both patterns arise from domain-specific assumptions about what constitutes randomness rather than from competing cognitive errors.

Recent work clarifies why such rapid shifts occur. Zhang et al. (2024) distinguish between top-down beliefs about randomness, which drive negative recency, and bottom-up sensitivity to local regularities, which supports positive recency. This distinction explains why identical streaks can elicit opposite predictions under minimal framing changes. Formal models reinforce this conclusion; Rabin and Vayanos (2010) show that biased learning about stochastic processes can generate both reversal and momentum expectations, depending on assumptions about how outcomes evolve over time. Taken together, the literature reviewed in this theme shows that apparent inconsistencies between the gambler's fallacy and hot-hand belief largely dissolve once context is taken into account. Judgments about streaks reflect competing inferential pressures: abstract beliefs about randomness versus concrete cues suggesting dependence. Theme 1 thus establishes the phenomenology of context dependence, setting the stage for a deeper examination of the mechanisms that generate these shifts.

Theme 2: Beliefs About Generative Structure as the Core Mechanism for Streak Predictions

Building on the contextual patterns identified in Theme 1, Theme 2 clarifies the underlying mechanism: people actively infer properties of the generative process producing outcomes, and these inferences determine whether streaks are interpreted as signaling reversal or continuation. Rather than responding directly to outcome sequences, individuals construct causal models that incorporate assumptions about independence, temporal dependence, latent states, and nonstationarity.

Evidence for this mechanism comes from studies showing that negative recency is strongest when outcomes are construed as independent and stationary, but weakens when cues imply dependence. Militana et al. (2010) demonstrate that longer intertrial intervals amplify gambler's fallacy effects, which is contrary to the previously established "replenishment hypothesis" that predicts that longer intervals allow more time for the random outcome to reset. This finding suggests that the bias is not solely due to the immediate, short-term memory of a sequence, but rather how individuals process and recall streaks over time. Furthermore, field evidence shows that such assumptions are not fixed. Muto et al. (2025) find

that baccarat gamblers frequently follow trends rather than expect reversals, even in objectively random environments, particularly when streaks are long and consecutive. These findings bolster that experiential cues and feedback can override default independence assumptions.

Framing studies provide further evidence that beliefs about structure can be manipulated without altering objective statistics. MacMahon et al. (2014) show that describing sequences in terms of momentum versus chance reliably shifts predictions toward continuation. Raab and MacMahon (2015) extend this effect to sports decision-making, demonstrating that reframing performance alters expectations even when underlying indicators remain unchanged. Together, these studies indicate that beliefs about dependence are not solely inferred from temporal structure, but can be imposed through conceptual schemas. Other work explicitly models how people infer latent structure from streaks. Asparouhova et al. (2009) and Rao and Hastie (2023) show that participants often assume nonstationary processes, inferring hidden regimes or state changes when observing runs of outcomes. Such inferences blend beliefs about randomness with assumptions of persistence, producing predictions that may resemble either gambler's fallacy or hot-hand behaviour depending on the inferred structure.

These beliefs have clear downstream consequences. Morais et al. (2025) demonstrate that hot-hand beliefs interact with loss aversion in financial decision-making, while Choi et al. (2025) show that encouraging holistic processing attenuates both positive and negative recency effects. Collectively, the literature converges on the conclusion that streak judgments are best understood as the application of inferred generative models, not as direct responses to surface patterns. Theme 2 thus identifies belief formation and application as the central mechanism underlying context-dependent streak predictions.

Theme 3: Agency, Intentionality, and Anthropomorphism

While Themes 1 and 2 establish that streak judgments depend on inferred generative structure, a critical question remains: what cues lead people to infer dependence, memory, or persistence in the first place? Theme 3 addresses this question by examining how attribution of agency and intentionality, often through anthropomorphism, systematically alters beliefs about how outcomes are generated.

A substantial body of work shows that anthropomorphizing nonhuman systems changes judgments under uncertainty by introducing assumptions of responsiveness, control, and internal state. In gambling contexts, Kim and McGill (2011) and Riva et al. (2015) demonstrate that anthropomorphized slot machines increase engagement and risk-taking despite identical payoff structures. Pancani et al. (2019) further show that social exclusion heightens anthropomorphism, which in turn amplifies gambling behaviour. Although these studies do not directly measure streak prediction, they converge on the implication that anthropomorphism increases perceived dependence across trials, precisely the belief shift identified in Theme 2 as a driver of hot-hand reasoning.

Other studies move beyond anthropomorphism to examine intentionality more directly. Caruso et al. (2010) show that when outcomes are attributed to intentional agents, observers judge streaks as more likely to continue, even when the underlying process is objectively random. Ericson et al. (2023) similarly argue that intentional action is assumed to reflect goals, memory, and learning, leading observers to expect persistence rather than regression. Braga and Jacinto (2024) extend these findings to organizational settings, showing that perceived intentionality amplifies managerial hot-hand predictions following strong performance. Together, these findings contrast sharply with judgments of mechanical or chance-based systems, where repetition more often triggers expectations of correction.

Importantly, agency attribution is not unitary. Kulow et al. (2021) demonstrate that anthropomorphized luck can induce perceptions of risk-sharing rather than competence, altering behaviour in ways that do not map cleanly onto hot-hand expectations. This heterogeneity suggests that different forms of agency attribution - competent versus capricious, fair versus manipulative - may generate opposing predictions about streak continuation or reversal. Thus, anthropomorphism could be understood as a higher-order cue that shapes inferred generative structure, rather than as a simple trigger for hot-hand belief.

Overall, these studies support the view that attributing agency fundamentally alters inferred generative structure by introducing assumptions of memory, learning, and internal state. These assumptions provide a coherent mechanism through which identical streaks can be interpreted as meaningful signals of momentum rather than random fluctuation. At the same time, the diversity of agency effects highlights a critical gap: few studies directly connect anthropomorphism to explicit streak prediction. This gap motivates the extensions developed in the discussion, where anthropomorphism is framed as a process-level moderator, and a promising direction for extending existing theories of sequential judgment.

Discussion

Across domains and methods, a clear pattern emerges: judgments about streaks are neither fixed biases nor simple misperceptions of probability, but context-sensitive inferences shaped by beliefs about the generative process. When outcomes are perceived as independent and self-correcting, negative recency dominates; when outcomes are interpreted as state-driven or intentional, expectations shift toward positive recency. This pattern helps reconcile long-standing debates over the gambler's fallacy and hot-hand belief, including conflicting empirical evidence about their prevalence and robustness (Sundali & Croson, 2006; Miller & Sanjurjo, 2018, 2024). Rather than asking which bias is "correct," the literature increasingly suggests that people flexibly deploy different inferential models depending on cues about randomness, control, and agency (Fischer & Savranevski, 2015; Reimers et al., 2018).

At the same time, important gaps and tensions remain. First, much of the literature infers beliefs about generative processes indirectly from behaviour, leaving open questions about the specific cognitive representations driving predictions. Second, methodological heterogeneity, ranging from abstract laboratory tasks to high-stakes field settings, contributes to apparent contradictions across findings (Chen et al., 2016; Dillon & Lybbert, 2024). While some studies emphasize robust negative recency in gambling contexts, others document strong hot-hand effects in sports, finance, and team-based decisions (Attali, 2013; Stöckl et al., 2015; Huber et al., 2010). These inconsistencies are difficult to resolve without explicit measurement of perceived dependence, intentionality, and state persistence. Individual differences further complicate this picture. Risk preferences, loss sensitivity, and prior experience modulate how people interpret streaks , suggesting that generative models are not only context-dependent but also person-dependent (Ball, 2012; Krébesz et al., 2023).

The present review highlights anthropomorphism and agency attribution as a promising extension for addressing these gaps. Prior work on social cognition and human-machine interaction shows that attributing agency to nonhuman systems alters trust, forgiveness, and expectations of future behaviour (Yuasa, 2020; Powdthavee & Riyanto, 2015). Integrating this insight with streak judgment research suggests two key directions for future work. First, anthropomorphism may systematically shift beliefs about generative structure by increasing perceived dependence and memory, thereby promoting hot-hand expectations. Second, anthropomorphism is not unitary: different forms of agency attribution (e.g., skilled versus inconsistent, fair versus manipulative) may generate opposing predictions about streak continuation or reversal. Testing these hypotheses would potentially move the field beyond debates over bias toward a process-level understanding of how people interpret sequences under uncertainty.

Conclusion

This review synthesized research on streak perception by integrating work on the gambler's fallacy, the hot-hand belief, and agency attribution. Across domains, the evidence converges on a central insight: people's judgments about streaks reflect coherent causal reasoning grounded in beliefs about how outcomes are generated, rather than simple errors in probability judgment. While substantial progress has been made in documenting when expectations of reversal versus continuation arise, less is known about the cues that shape inferred generative structure. By positioning agency attribution,

and anthropomorphism in particular, as higher-order cues that influence assumptions of dependence and memory, this review identifies a promising direction for extending existing models of sequential judgment.

Understanding how people interpret streaks has implications beyond gambling and sports. As humans increasingly interact with algorithmic systems, AI agents, and complex automated environments, assumptions about agency and intent are likely to play a growing role in decision-making under uncertainty. A process-level account of streak judgment that integrates generative inference and agency attribution is therefore beneficial for both theory and application.

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